

# ELI — Diagrams

ELI v2.3.17

August 2026

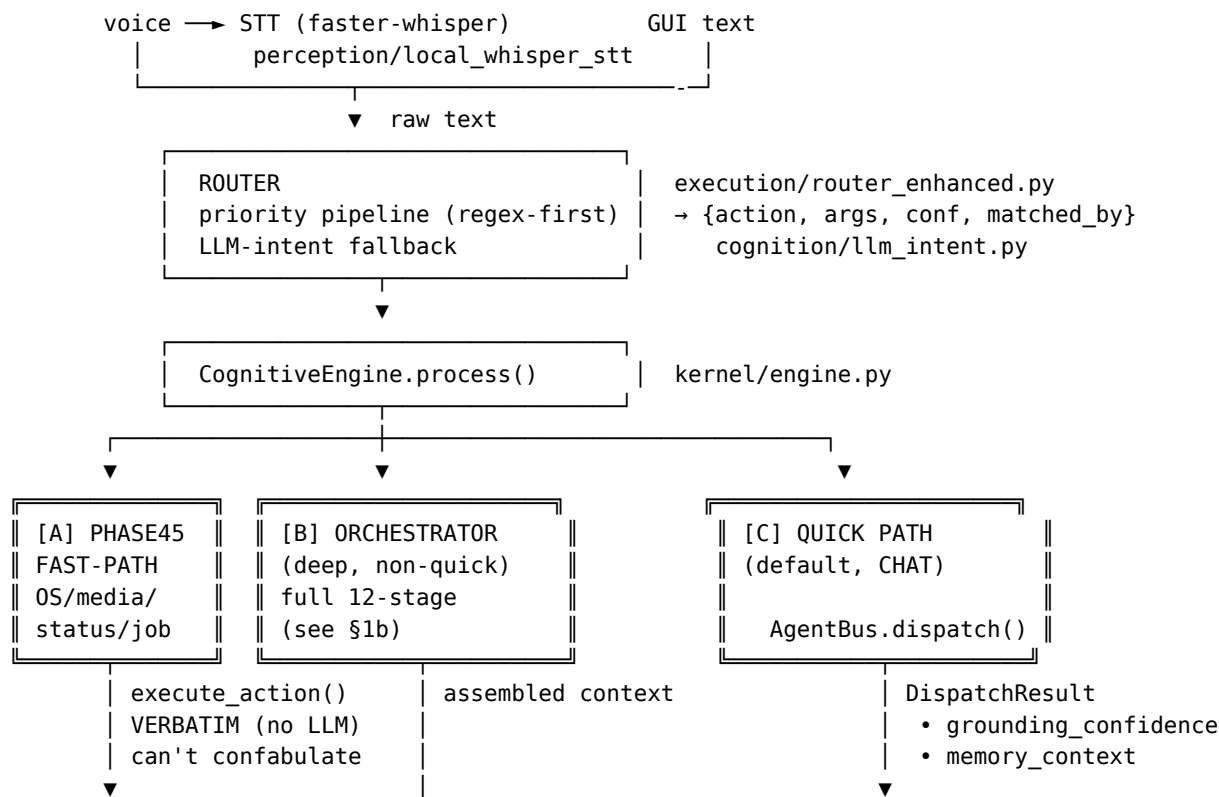
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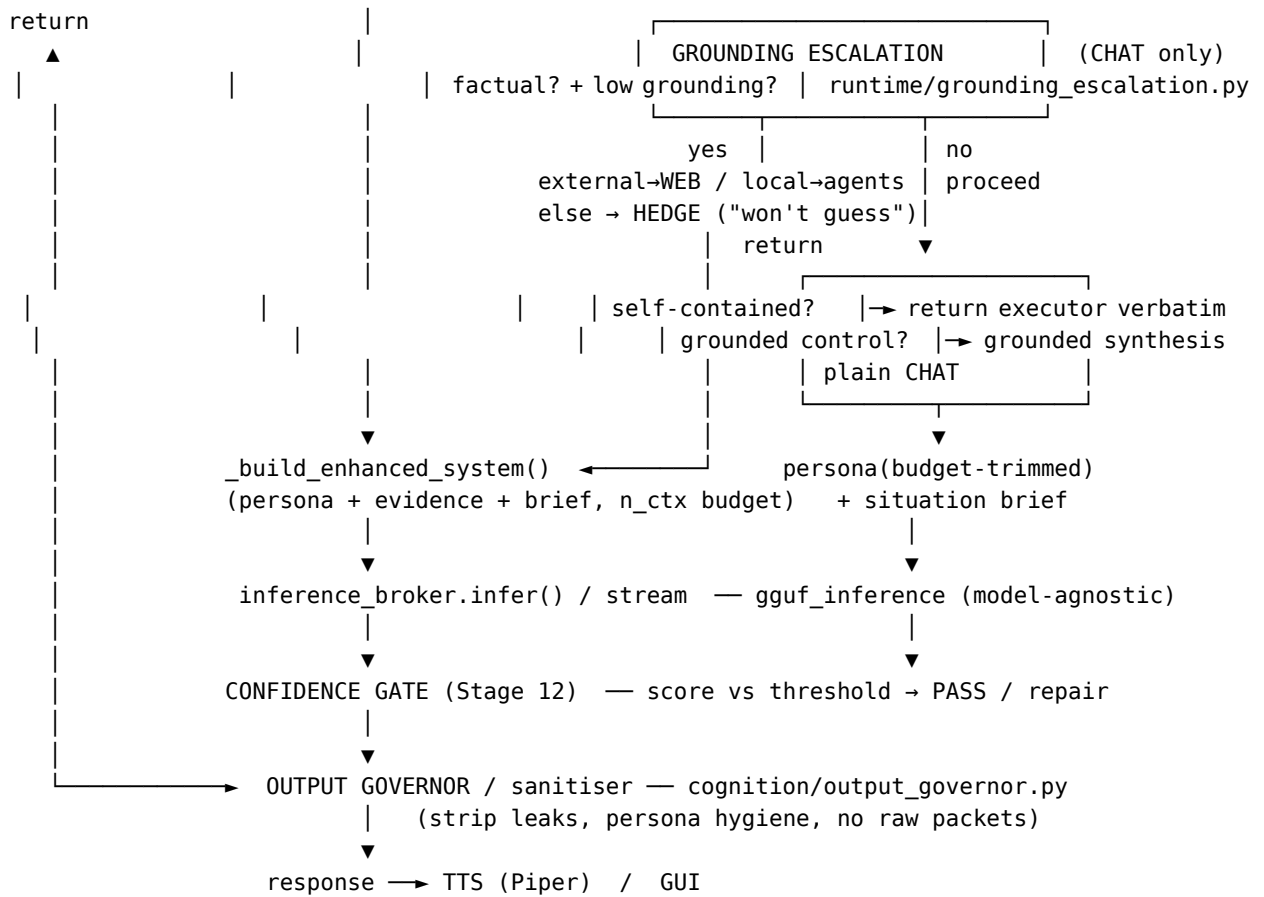
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## Blueprint — ELI MKXI ASCII Diagrams

Visual companion to architecture.md. Three views: the full request pipeline, the memory subsystem, and the gating stack. All grounded in the real modules (file paths are clickable).

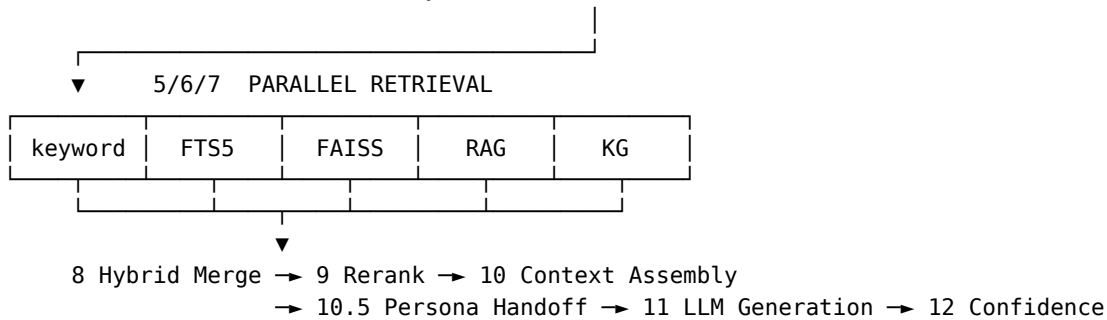
### 1. Full pipeline (request lifecycle)



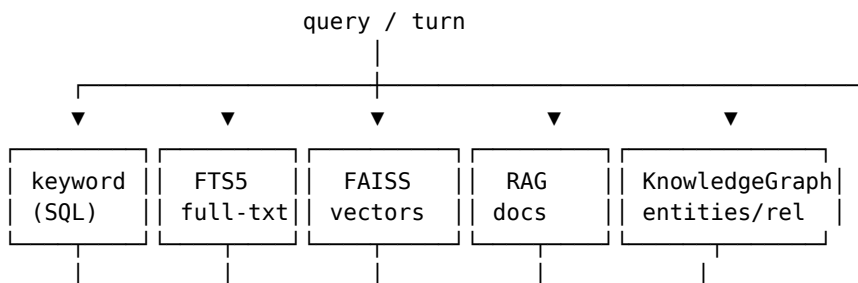


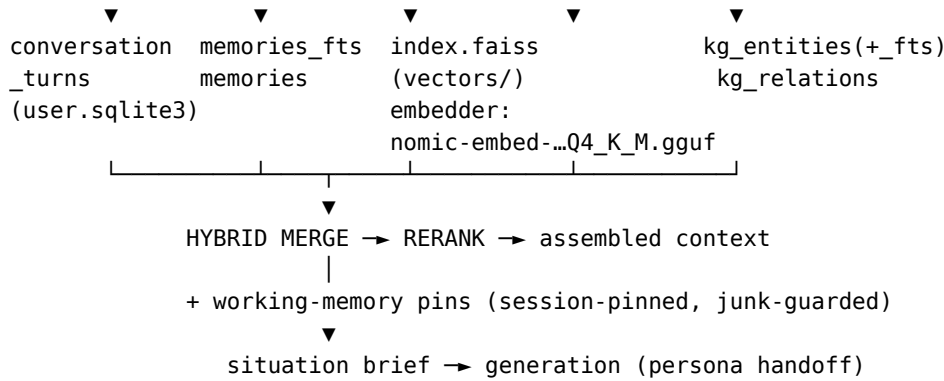
## 1b. Orchestrator — the 12 stages (path [B]) cognition/orchestrator.py

1 Intent → 2 Persona Lock → 3 HyDE → 4 Planner



## 2. Memory subsystem eli/memory + cognition/working\_memory.py





| STORES (artifacts/)                  |                                                                                                                                                                                                                                                                                                     |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| db/user.sqlite3                      | memories(+fts), conversation_turns, conversations, kg_entities(+fts), kg_relations, recall_log, runtime_events, news_articles(+fts), news_reflections, habits/habit_events/habit_rules, observations, learning_replay, working_memory_pins, user_patterns, session_summaries, corrections, failures |
| db/agent.sqlite3                     | agent_dispatches, agent_metrics, improvements, failures, code_patches, error_tracking, observations                                                                                                                                                                                                 |
| vectors/index.faiss                  | semantic index                                                                                                                                                                                                                                                                                      |
| runtime/users/<uuid>/user_profile... | per-user profile                                                                                                                                                                                                                                                                                    |

WRITE PATH: turn → PERSISTENCE GATE (drop junk/report-dumps) → store

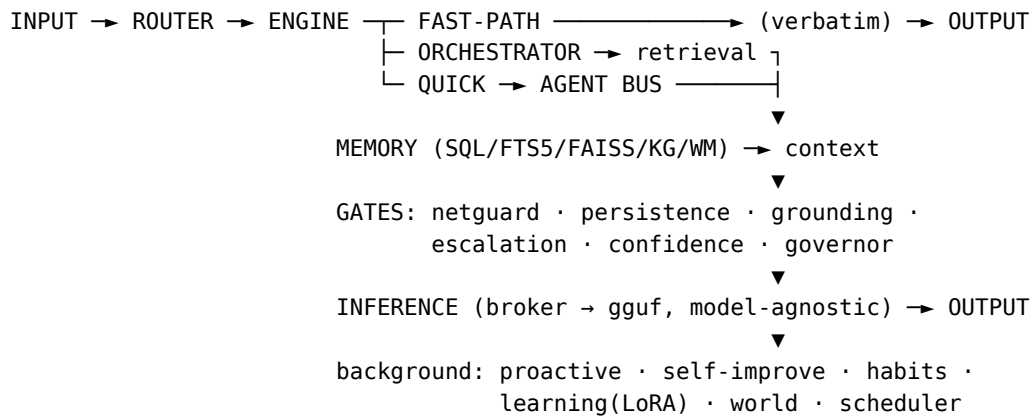
### 3. Gating stack (in path order — every guard the request passes)

|   |                                                                       |                              |  |
|---|-----------------------------------------------------------------------|------------------------------|--|
| 1 | NETGUARD (process-wide, always-on)                                    | core/netguard.py             |  |
|   | any outbound socket → offline? → OfflineError (FAIL-CLOSED)           |                              |  |
|   | allow_network(): scoped opt-in window (model download, web tier)      |                              |  |
| 2 | PHASE45 FAST-PATH GATE                                                | kernel/engine.py             |  |
|   | deterministic action (volume/media/date/job/status)?                  |                              |  |
|   | → execute_action() VERBATIM, no LLM (cannot confabulate)              |                              |  |
| 3 | PERSISTENCE GATE                                                      | runtime/persistence_gate.py  |  |
|   | about to write to memory? junk / internal report-dump?                |                              |  |
|   | → DROP (so it can't be stored and replayed as fact later)             |                              |  |
| 4 | GROUNDING GATE                                                        | runtime/deterministic_gate   |  |
|   | answer requires evidence? unbacked or raw packet?                     |                              |  |
|   | → BLOCK raw evidence/telemetry leak; require grounded synthesis       |                              |  |
| 5 | GROUNDING ESCALATION                                                  | runtime/grounding_escalation |  |
|   | checkable fact AND grounding < threshold ?                            |                              |  |
|   | external fact → WEB tier                                              |                              |  |
|   | self/project → LOCAL tier                                             |                              |  |
|   | exhausted / offline                                                   |                              |  |
|   | first tier that grounds → answer                                      |                              |  |
|   | → HEDGE ("I won't guess")                                             |                              |  |
|   | (trigger = GROUNDING, not the response score – which lies when wrong) |                              |  |

|   |   |                                                               |                              |  |
|---|---|---------------------------------------------------------------|------------------------------|--|
| 6 | ⊞ | CONFIDENCE GATE (Stage 12)                                    | kernel/engine.py             |  |
|   |   | response score vs threshold → PASS / repair pass              |                              |  |
| 7 | ⊞ | OUTPUT GOVERNOR / SANITISER                                   | cognition/output_governor.py |  |
|   |   | final text → strip leaks, persona hygiene, no raw JSON → user |                              |  |

Legend: FAIL-CLOSED = denies by default; VERBATIM = returned without an LLM pass; HEDGE = honest "can't verify" instead of a guess.

#### 4. One-screen overview (how it all connects)



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